



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A Place to Learn, A Chance to Grow"

Session: 2025-26



VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	CS6T002	Software Testing & Quality Assurance	2	0	0	2

Prerequisites for the course

1.	Basic understanding of software engineering principles .
2.	Basic experience with some test automation tools .

Prior Reading Material/useful links

1.	https://www.geeksforgeeks.org/software-testing-vs-quality-assurance/
2.	https://www.tutorialspoint.com/software_testing/software_testing_qa_qc_testing.htm

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will be able to Understanding the Basics of Software Testing and Its Importance.
2	CO2	Student will be able to gain a thorough understanding of Unit Testing and Debugging Techniques.
3	CO3	Student will be able to Identify and evaluate different paths in a Control Flow Graph using appropriate path selection criteria.
4	CO4	Student will be able to Construct and interpret Data Flow Graphs to analyze the flow of data within a program.
5	CO5	Student will be able to Explain different types of software testing, including functional and performance testing techniques.

Syllabus:

Course Contents		Hours
Unit I	Basic concepts of Testing: Need of Testing, Basic concepts- errors, faults, defects, failures, objective of testing, central issue in testing, Testing activities, V-Model, Monitoring and Measuring Test Execution, Test tools and Automation, Limitation of Testing.	07
Unit II	Unit Testing: Concepts of Unit Testing, Static Unit Testing, Defect Prevention, Dynamic Unit Testing, Mutation Testing, Debugging, Tools for Unit Testing.	07
Unit III	Control Flow: Control Flow Testing, Control Flow Graphs, Path in Control Flow Graph, Path Selection Criteria, Generating Test Input and Data Selection Examples.	07



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Unit IV	Data Flow Testing: Introduction to Data Flow Testing, Data Flow Graphs, Data Flow Testing Criteria, Comparison of Data Flow Test Selection Criteria.	05
Unit V	Software Testing Categories: Software testing Types, Functionality test, Performance test, Load Test, Stress test, and Scalability test.	04

Text Books

1.	Software Testing and Quality Assurance by Kshirsager Naik and Priyadarshini Tripathi (Wiley)
2.	Software Testing Concepts and Tools by Nageswara Rao Pusuluri (Dream Tech Press).
3.	Software Testing Principles, Techniques and tools, 1st Edition, by M. G. Limaye McGraw Hills.

Reference Books

1.	"Foundations of Software Testing" 2E by Aditya P. Mathur , Pearson Education.
2.	Effective Methods for Software Testing- William E Perry, (Wiley). 2. Software Testing Tools by Dr. K. V. K. K. Prasad (Dream Tech)

Useful Links

1.	https://www.intertek.com/software/quality-assurance/
2.	https://qalified.com/blog/what-is-qa-testing/
3.	https://testsigma.com/blog/software-testing-and-quality-assurance/